$k\vec{A} = ka_1\hat{i} + ka_2\hat{j} + ka_3\hat{k}$

For $k \in \mathbb{R}, k > 1$ yields magnification, $0 < k < 1$ yields contraction of a vector and $k = 0$ yields a zero vector $\overrightarrow{OA} = 0\hat{i} + 0\hat{j} + 0\hat{k} = \vec{0}$.

Result 8.10

Using the commutative, associative properties of vector addition and the distributive property of the scalar multiplication we can prove the following.

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and if m is a scalar, then

(i) $\vec{a} + \vec{b} = (a_1 + b_1)\hat{i} + (a_2 + b_2)\hat{j} + (a_3 + b_3)\hat{k}$

(ii) $\vec{a} - \vec{b} = (a_1 - b_1)\hat{i} + (a_2 - b_2)\hat{j} + (a_3 - b_3)\hat{k}$

(iii) $m\vec{a} = ma_1\hat{i} + ma_2\hat{j} + ma_3\hat{k}$ and

(iv) $\vec{a} = \vec{b}$ if and only if $a_1 = b_1$, $a_2 = b_2$, and $a_3 = b_3$.

Example 8.4

Find a unit vector along the direction of the vector $5\hat{i} - 3\hat{j} + 4\hat{k}$.

Solution

We know that a unit vector along the direction of the vector $\vec{a}$ is given by $\frac{\vec{a}}{|\vec{a}|}$. So a unit vector along the direction of $5\hat{i} - 3\hat{j} + 4\hat{k}$ is given by

$$\frac{5\hat{i} - 3\hat{j} + 4\hat{k}}{|5\hat{i} - 3\hat{j} + 4\hat{k}|} = \frac{5\hat{i} - 3\hat{j} + 4\hat{k}}{\sqrt{5^2 + 3^2 + 4^2}} = \frac{5\hat{i} - 3\hat{j} + 4\hat{k}}{\sqrt{50}}.$$

Note 8.3

Now we have another unit vector parallel to $5\hat{i} - 3\hat{j} + 4\hat{k}$ in the opposite direction. That is,

$$-\frac{5\hat{i} - 3\hat{j} + 4\hat{k}}{\sqrt{50}}.$$

8.7 Direction Cosines and Direction Ratios

Let P be a point in the space with coordinates (x, y, z) and of distance r from the origin. Let R, S and T be the foots of the perpendiculars drawn from P to the x, y and z axes respectively. Then

$$\angle PRO = \angle PSO = \angle PTO = 90°.$$
$$OR = x,\ OS = y,\ OT = z \text{ and } OP = r.$$

(It may be difficult to visualize that $\angle PRO = \angle PSO = \angle PTO = 90°$ in the figure; as they are foot of the perpendiculars to the axes from P; in a three dimensional model we can easily visualize the fact.)

Let α, β, γ be the angles made by the vector $\overrightarrow{OP}$ with the positive x, y and z axes respectively. That is,

$$\angle POR = \alpha,\ \angle POS = \beta,\ \text{and}\ \angle POT = \gamma.$$

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